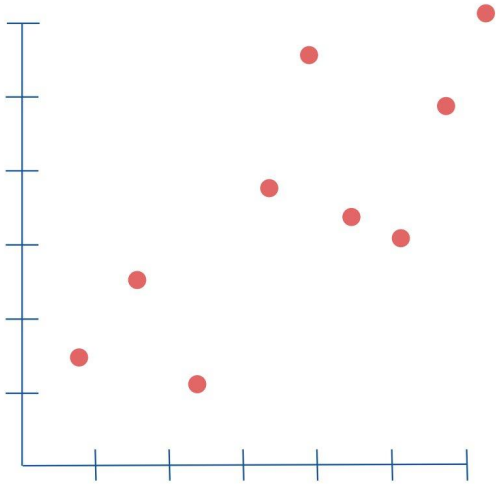
Aprendizaje Automático II

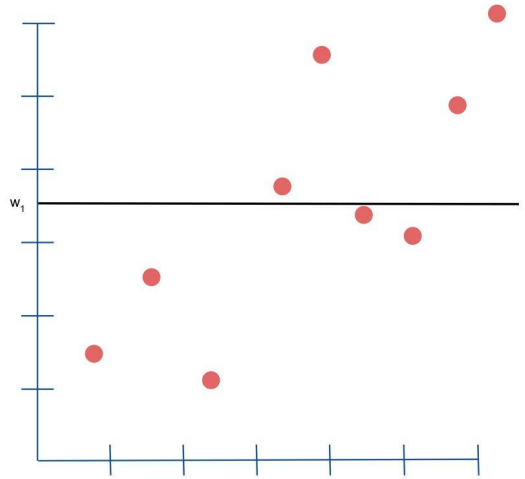
Dr. Federico Albanese
Franco Castellacci

Introducción a las Redes Neuronales

Regresión Lineal

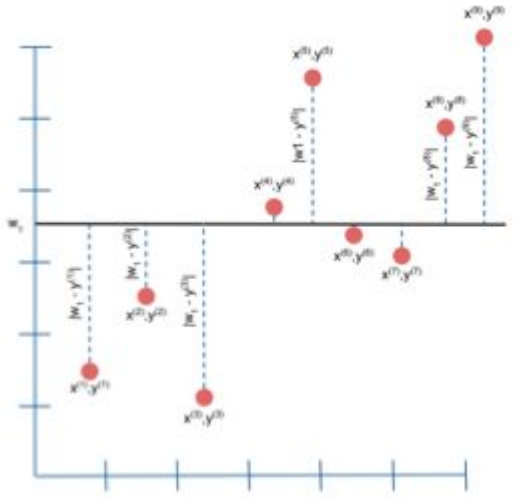


Regresión Lineal



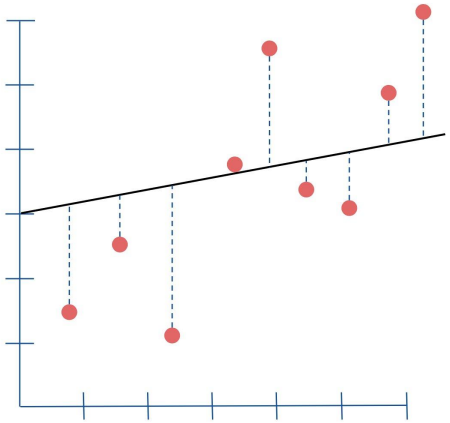
Mi “modelo” es una lineal.

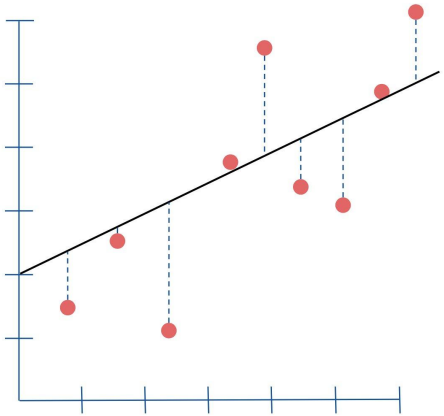
Regresión Lineal

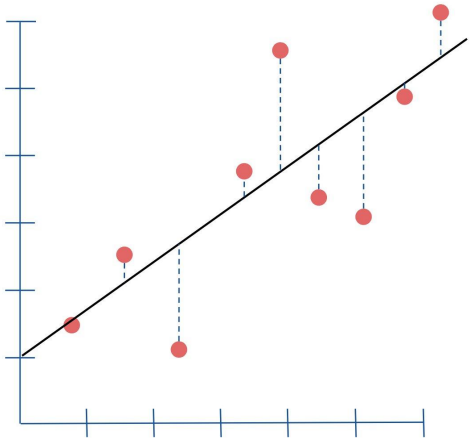


Para encontrar el mejor “modelo”
debo poder cuantificar el error y minimizarlo

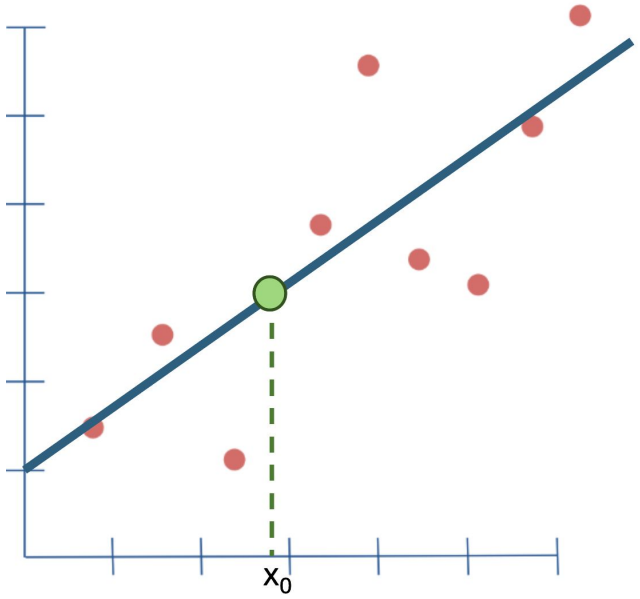
Regresión Lineal





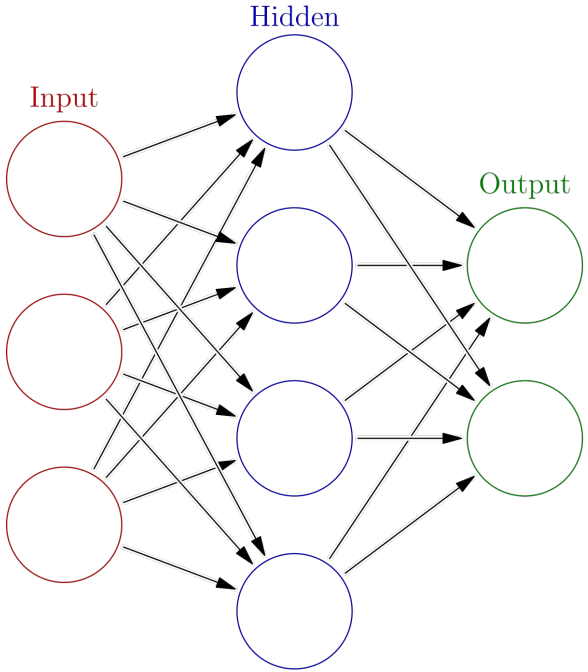


Regresión Lineal

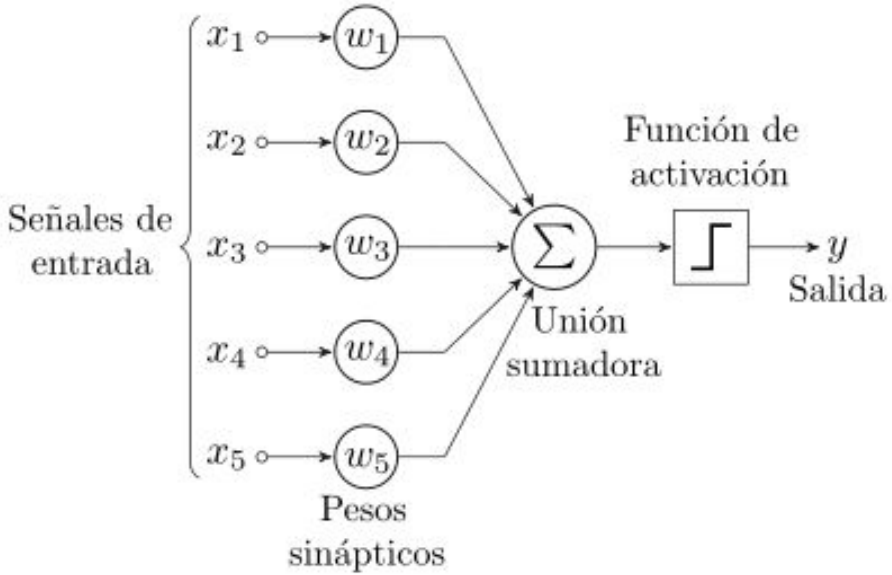


A partir de mi modelo generado con datos puedo inferir o “predecir”

Red neuronal

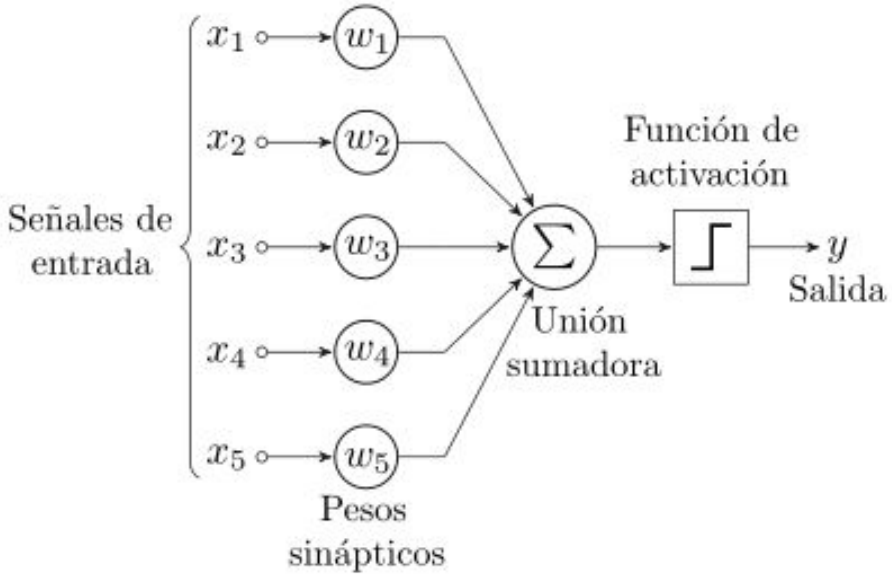


Perceptrón simple



$$Y = \sigma(\Sigma W_i * X_i + b)$$

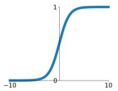
Perceptrón simple



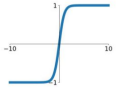
$$Y = \sigma(\Sigma W_i * X_i + b)$$

Activation Functions

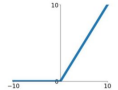
Sigmoid
$\sigma(x) = \frac{1}{1+e^{-x}}$



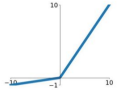
tanh
$\tanh(x)$



ReLU
$\max(0, x)$

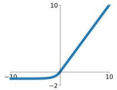


Leaky ReLU
$\max(0.1x, x)$



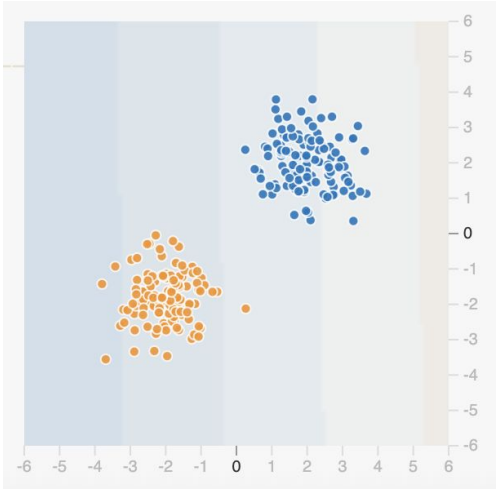
Maxout
$\max(w_1^T x + b_1, w_2^T x + b_2)$

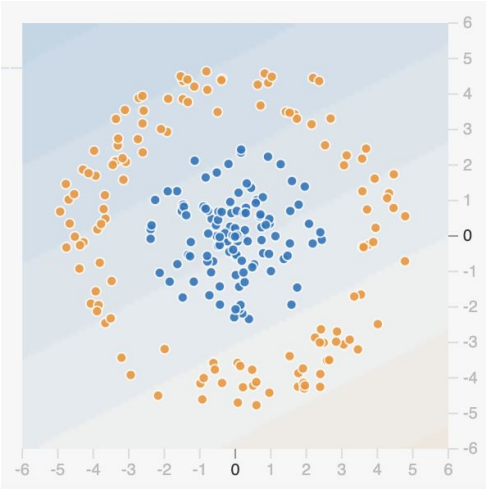
ELU
$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



¿Funciones de Activación? ¿Para qué?

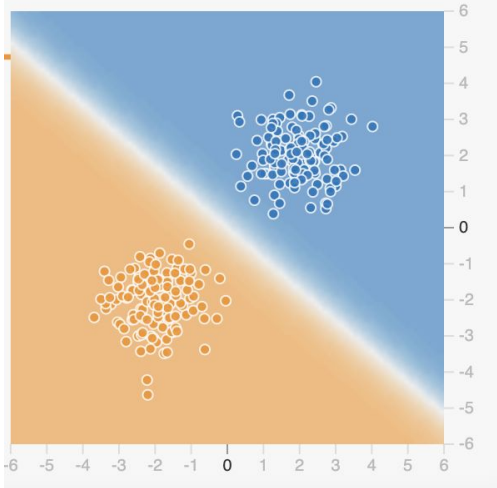
Las funciones de activación permiten introducir no-liniaridades a la red neuronal

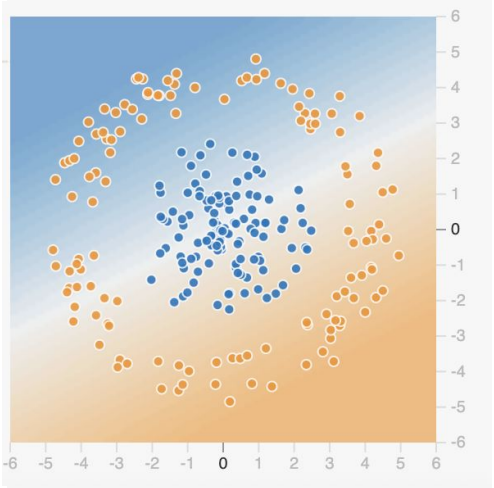




¿Funciones de Activación? ¿Para qué?

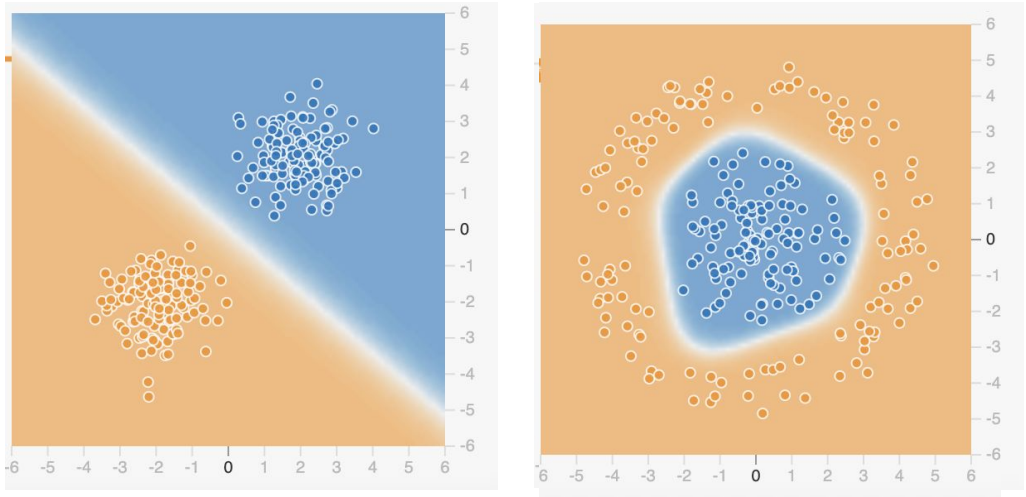
¿Qué pasa si quiero ajustar una función no lineal?



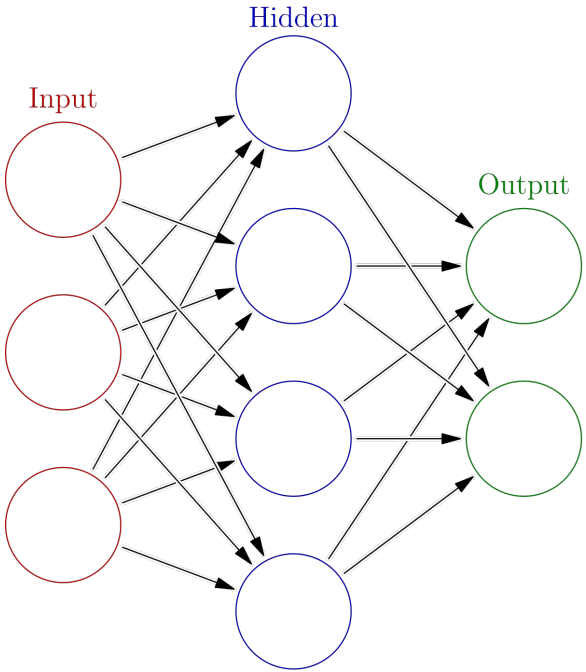


¿Funciones de Activación? ¿Para qué?

Necesito introducir no-liniaridades a la red neuronal



Red Neuronal



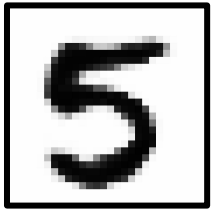
Una Red Neuronal solo son muchos perceptrones simples conectados entre sí.

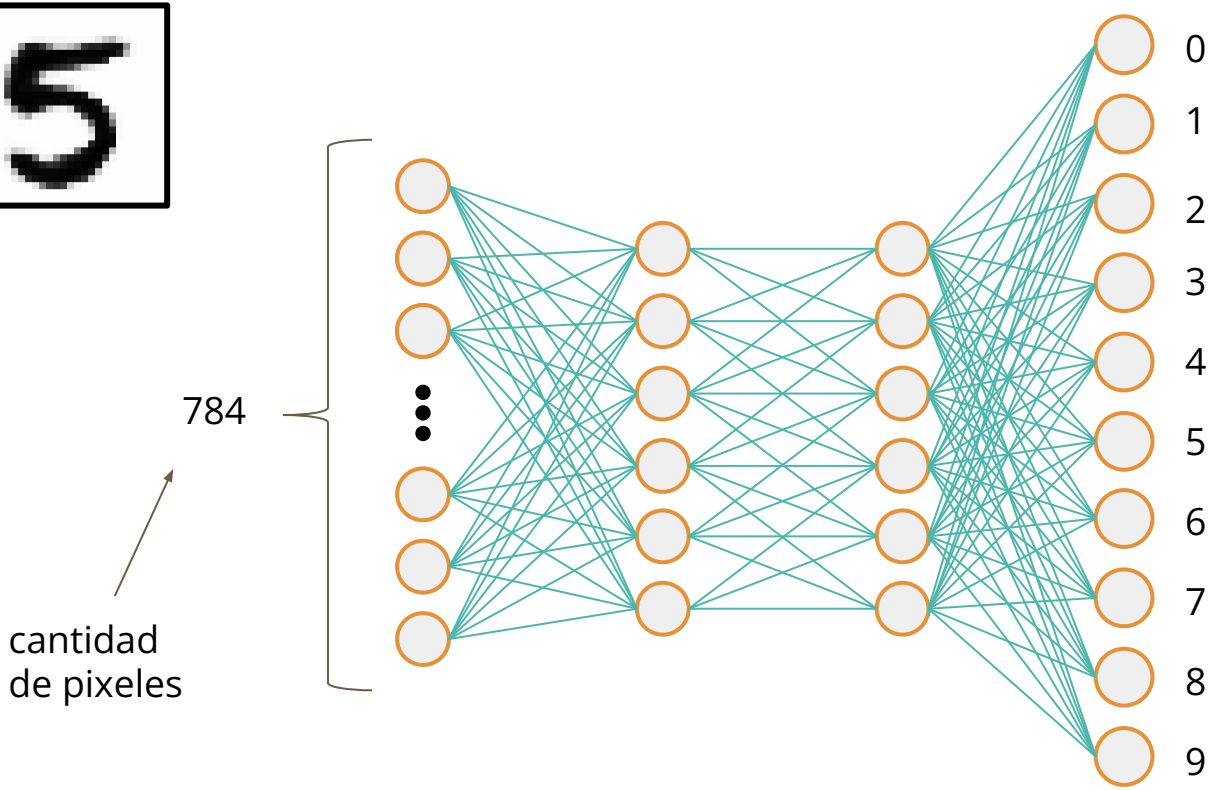
Red Neuronal

Ejemplo: mnist

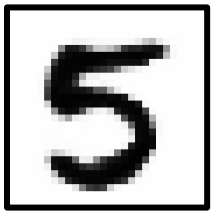


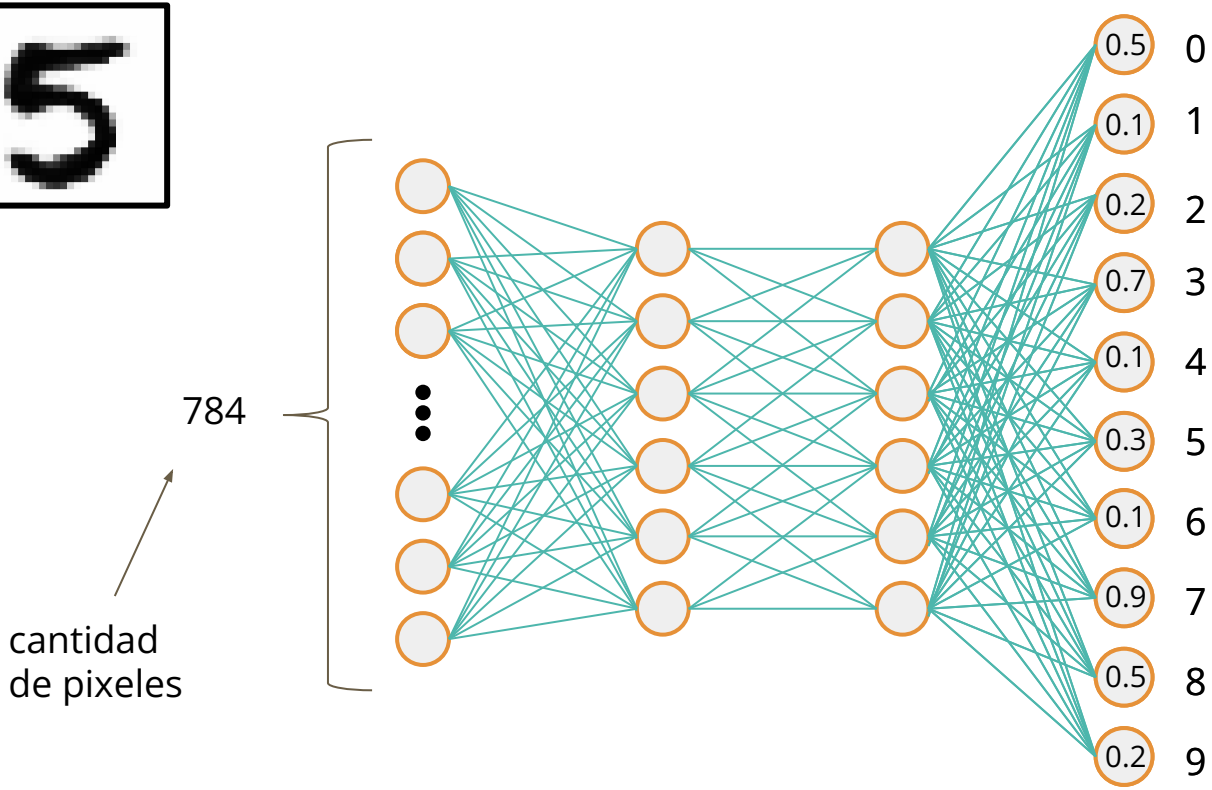
Red Neuronal: Backpropagation o cómo calcular w y b



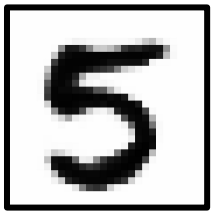


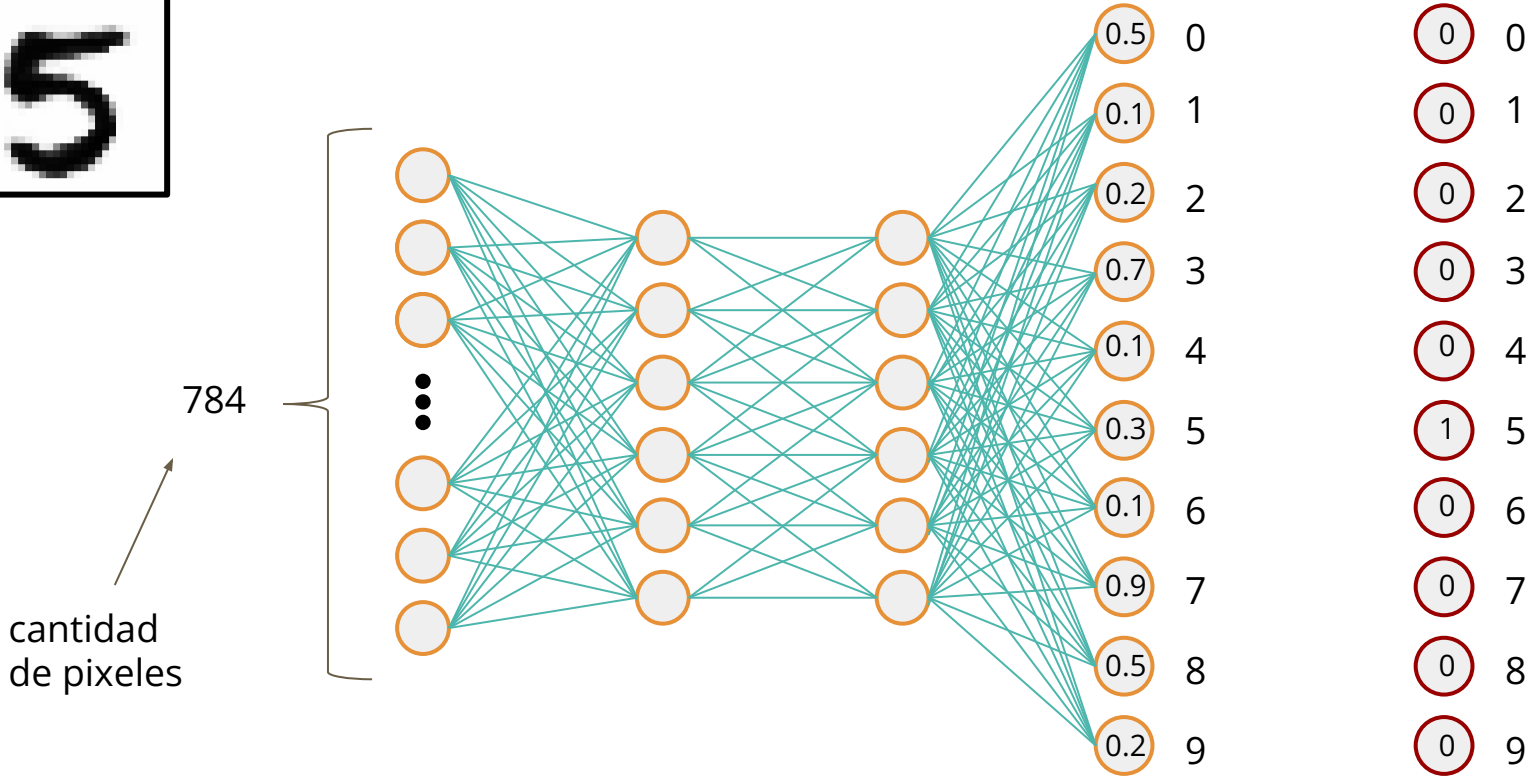
Red Neuronal: Backpropagation o cómo calcular w y b



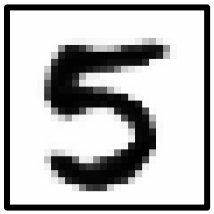


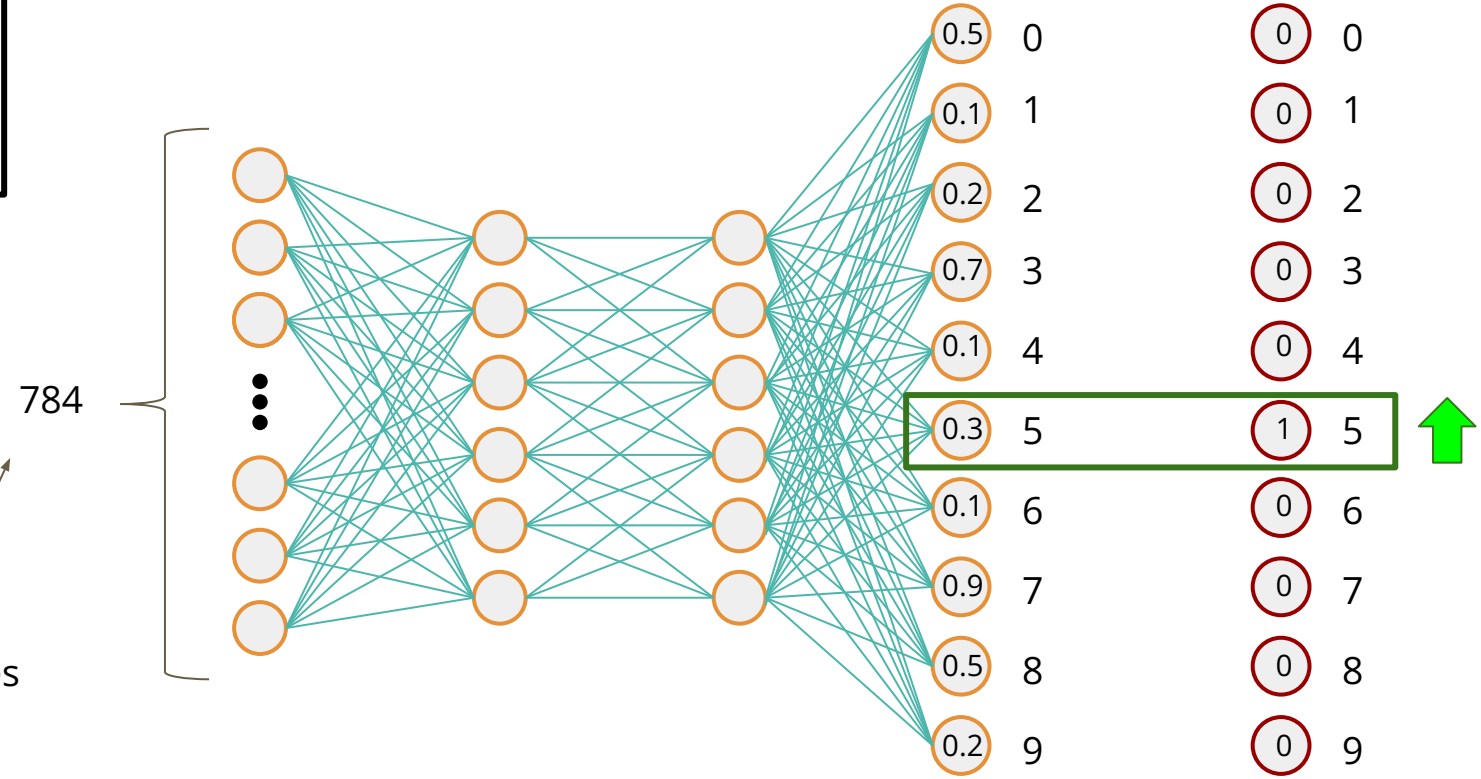
Red Neuronal: Backpropagation o cómo calcular w y b



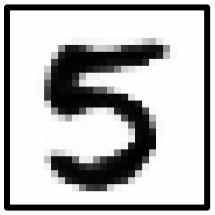


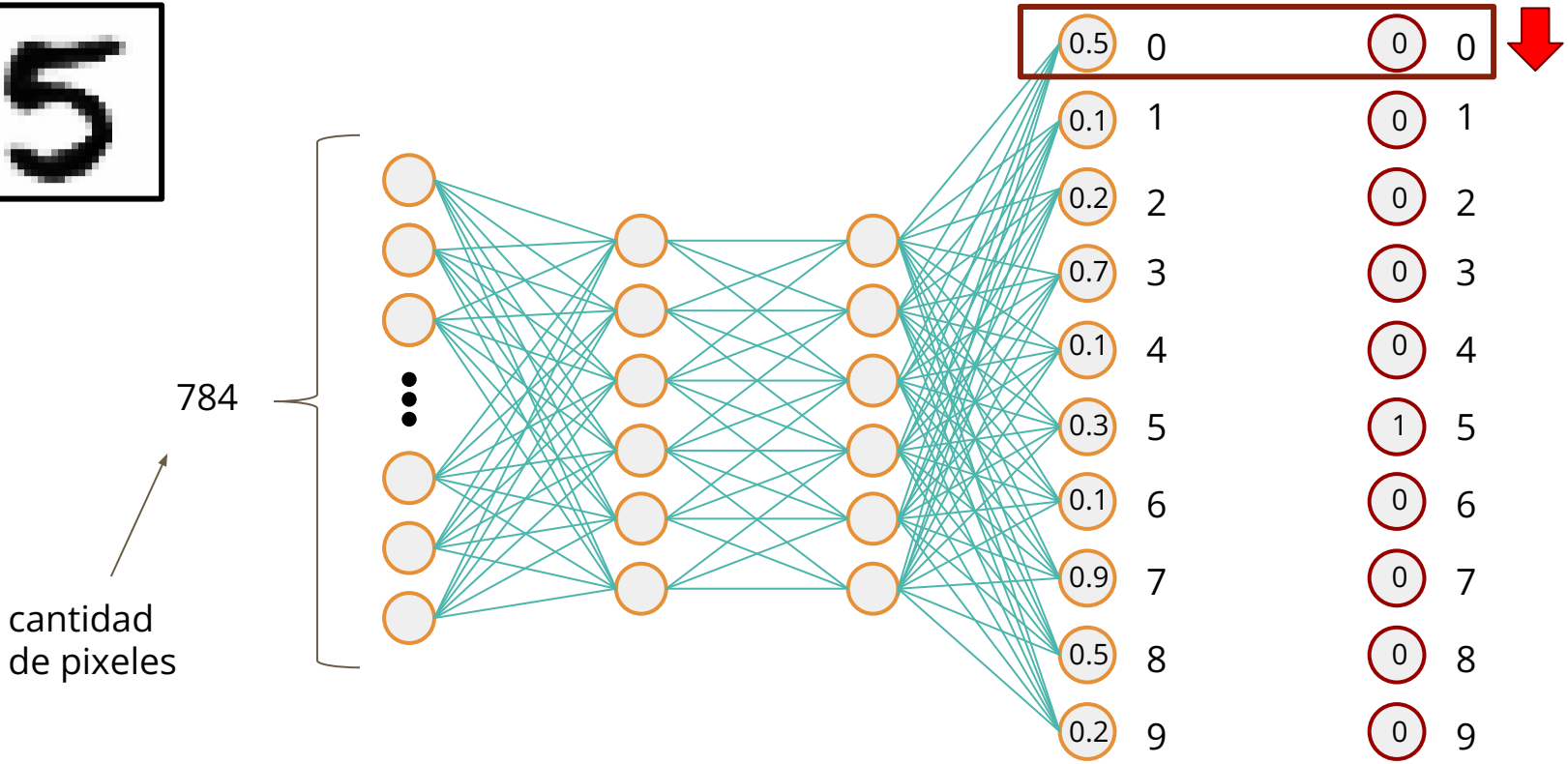
Red Neuronal: Backpropagation o cómo calcular w y b



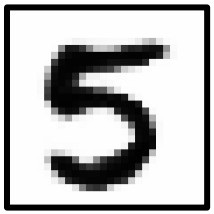


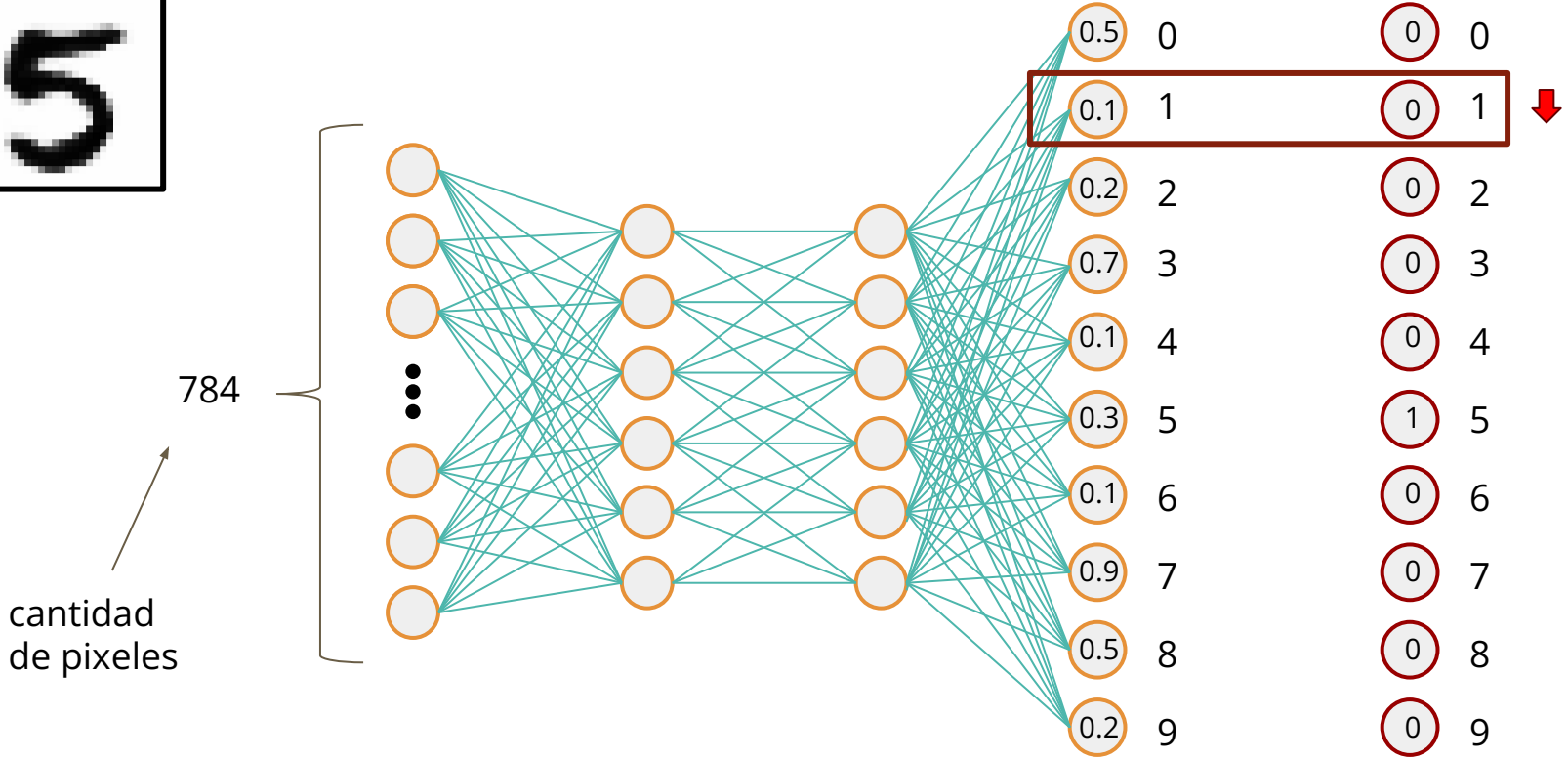
Red Neuronal: Backpropagation o cómo calcular w y b





Red Neuronal: Backpropagation o cómo calcular w y b





Red Neuronal: Backpropagation

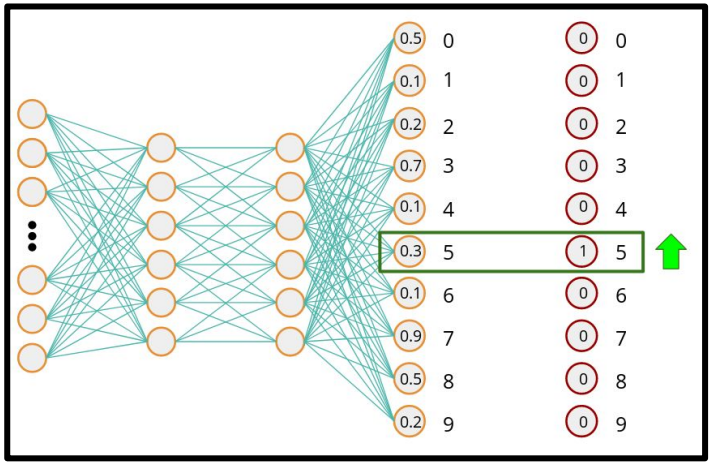
Se define una función de costo que determina mi error a partir del valor predicho y el valor real.
Por ejemplo la función de pérdida cuadrática:

$$Cost = (f(x) - Y)^2$$

Donde el valor predicho f(x) es:

$$f(x) = \sigma(z^l)$$

$$z^l = \sum_i w_i^l * a_i^{l-1} + b^l$$



Donde a_i^{l-1} corresponde al valor que toma la neurona i de la capa l-1.

Red Neuronal: Backpropagation

Para saber cuánto tengo que cambiar los valores, calculo cómo variaciones en w modifican la función de costo:

$$\frac{\partial C_0}{\partial w^l} = \frac{\partial z^l}{\partial w^l} * \frac{\partial a^l}{\partial z^l} * \frac{\partial C_0}{\partial a^l} = a^{l-1} * \sigma'(z^l) * 2(a^l - y)$$

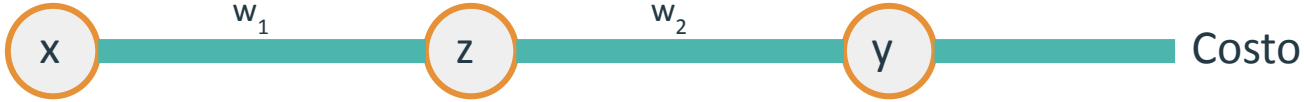
Luego:

$$\frac{\partial C}{\partial w^l} = \frac{1}{n} \sum_{k=0}^{n-1} \frac{\partial C_k}{\partial w^l}$$

Equivalente para el bias.

Red Neuronal: Backpropagation

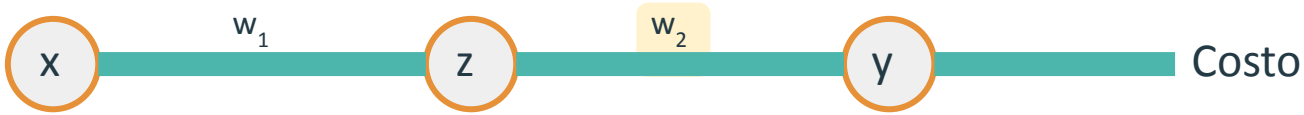
Ejemplo simple de backpropagation



$$\frac{\partial C}{\partial w_2} = ?$$

Red Neuronal: Backpropagation

Ejemplo simple de backpropagation



$$\frac{\partial C}{\partial w_2} = ?$$

Red Neuronal: Backpropagation

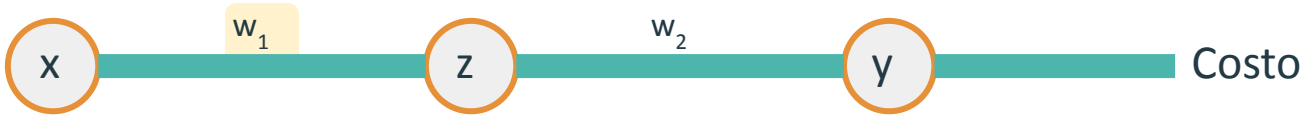
Ejemplo simple de backpropagation



$$\frac{\partial C}{\partial w_2} = \frac{\partial C}{\partial y}\frac{\partial y}{\partial w_2}$$

Red Neuronal: Backpropagation

Ejemplo simple de backpropagation



$$\frac{\partial C}{\partial w_1} = \frac{\partial C}{\partial y}\frac{\partial y}{\partial w_1}$$

Red Neuronal: Backpropagation

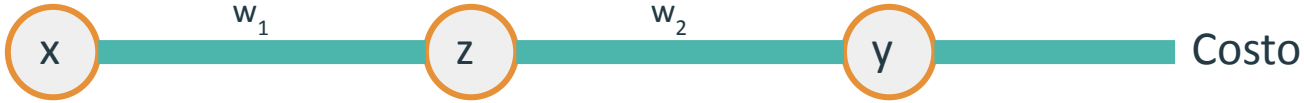
Ejemplo simple de backpropagation



$$\frac{\partial C}{\partial w_1} = \frac{\partial C}{\partial y}\frac{\partial y}{\partial z}\frac{\partial z}{\partial w_1}$$

Red Neuronal: Backpropagation

Ejemplo simple de backpropagation



$$\frac{\partial C}{\partial w_1} = \frac{\partial C}{\partial y}\frac{\partial y}{\partial z}\frac{\partial z}{\partial w_1}$$

Repetir esto para todos los pesos de la red

Red Neuronal: Backpropagation

Finalmente, para actualizar los valores de w hago:

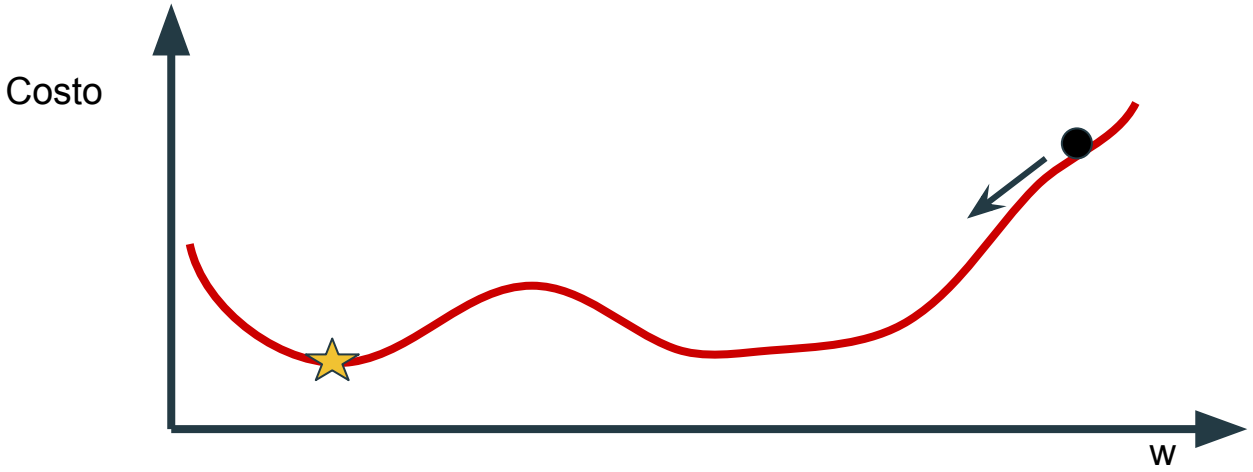
$$W_i^+ = W_i - \eta * \frac{\partial C}{\partial w_i}$$

donde η es el “learning rate”.

Learning rate

¿Pero cómo defino el valor del “learning rate”?

$$W_i^+ = W_i - \eta * \frac{\partial C}{\partial w_i}$$



Learning rate

¿Pero cómo defino el valor del “learning rate”?

Alternativas:

- Pruebo varios valores.

Learning rate

¿Pero cómo defino el valor del “learning rate”?

Alternativas:

- Pruebo varios valores.
- Voy adaptando el valor.

Learning rate

¿Pero cómo defino el valor del “learning rate”?

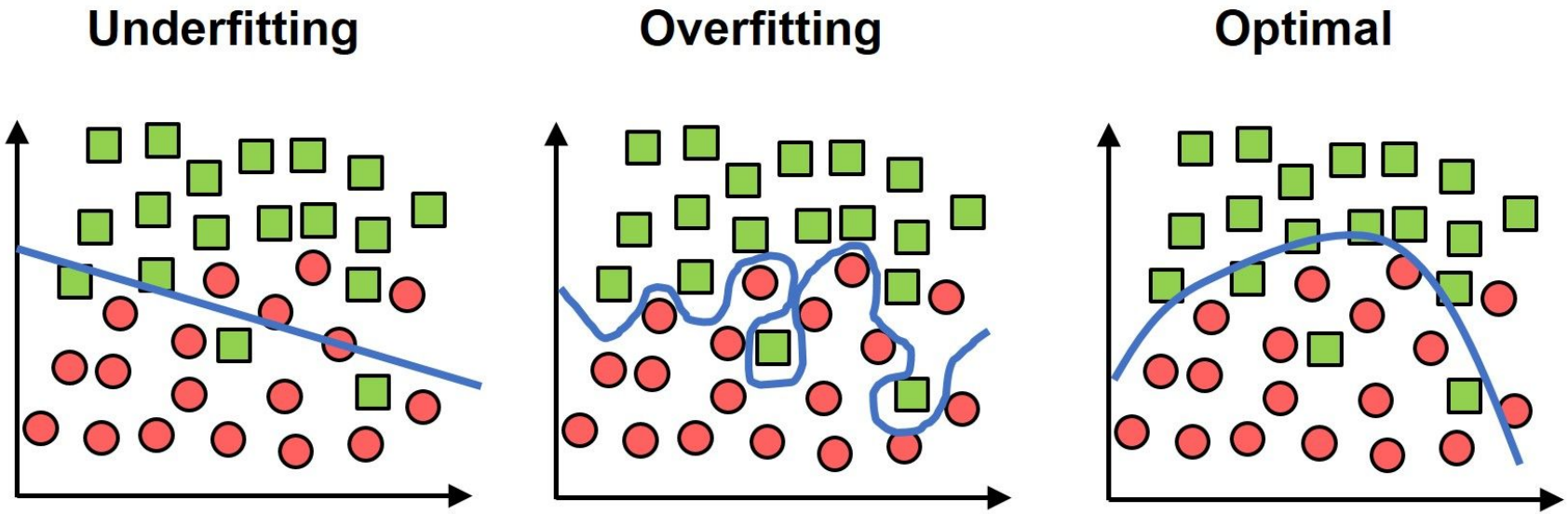
Alternativas:

- Pruebo varios valores.

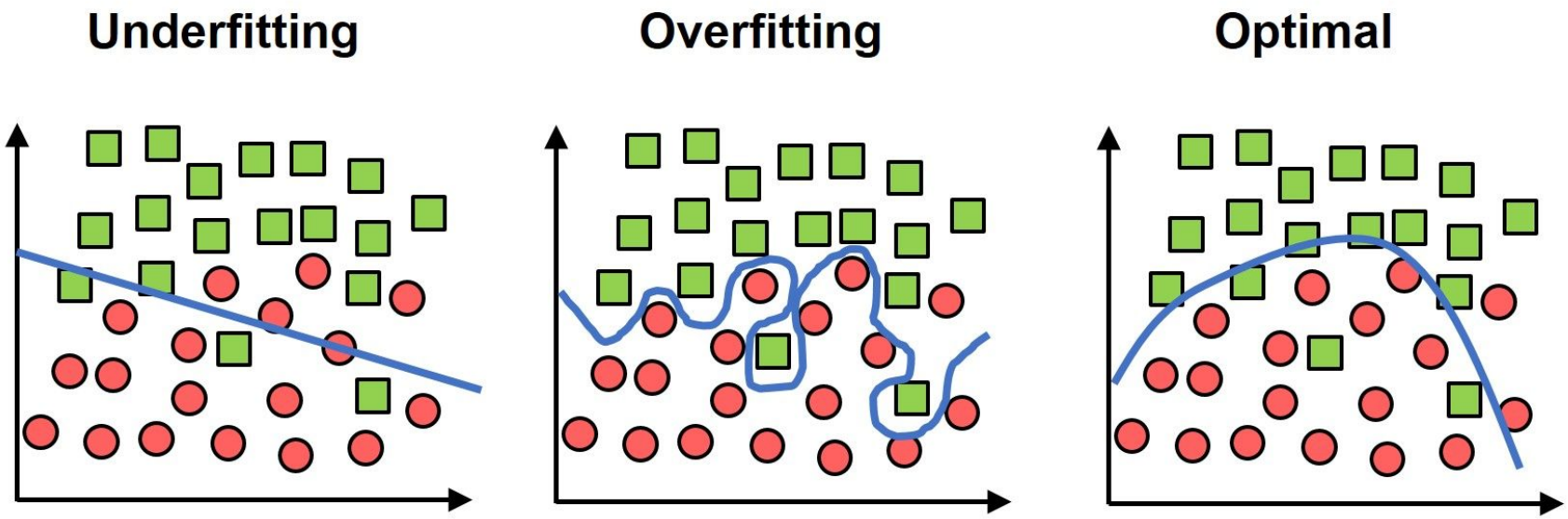
- Voy adaptando el valor.
  - Dependiendo del gradiente
  - El tamaño de los pesos
  - La iteración
  - …

Regularización

Regularización



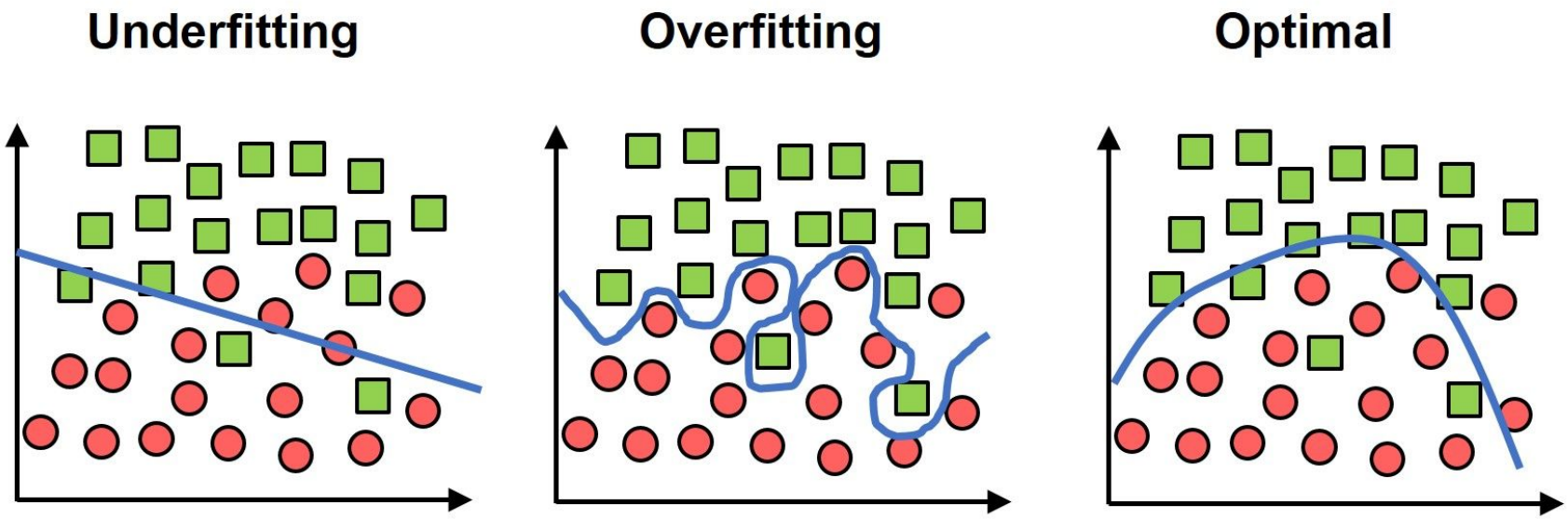
Regularización



“Pensar es olvidar diferencias, es generalizar, abstraer.
En el abarrotado mundo de Funes no había sino detalles”

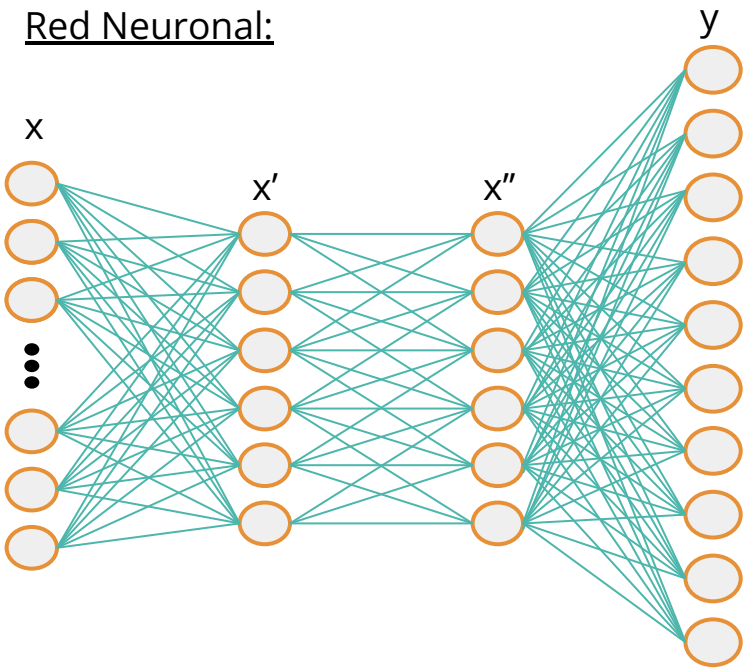


Regularización

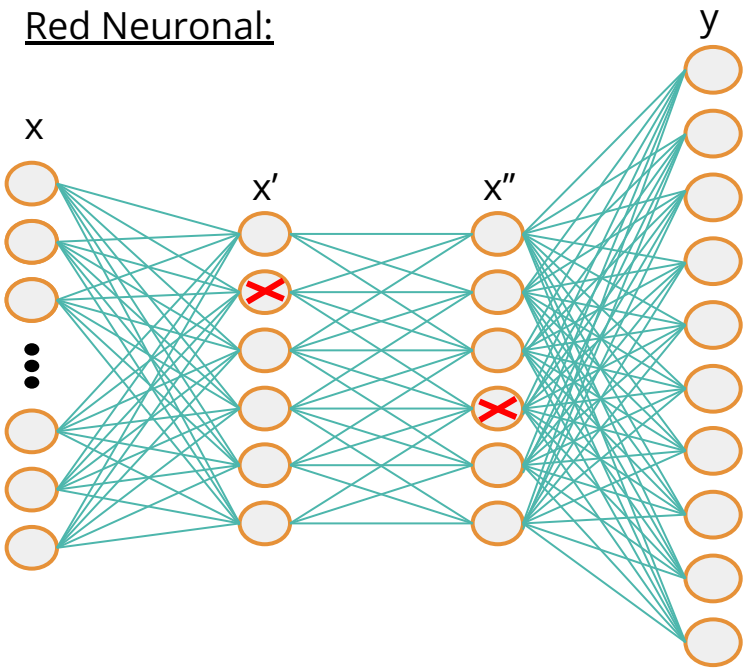


Objetivo de la regularización:
Mejorar la capacidad de generalización del modelo

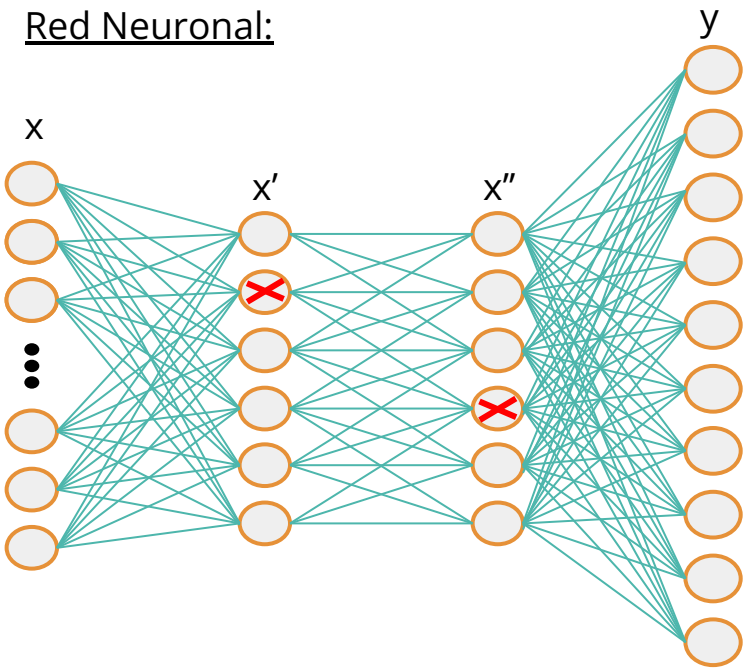
Regularización: Dropout



Regularización: Dropout



Regularización: Dropout



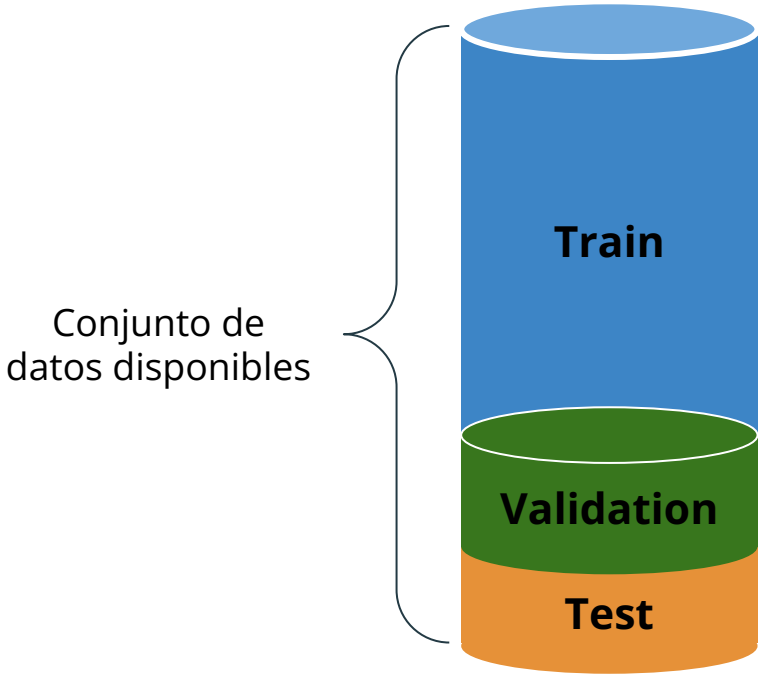
Formalmente:

- Multiplicamos por un vector binario (de 1 y 0) aleatorio para apagar algunas neuronas.
- Se define una probabilidad p de que una neurona se apague o no.

Regularización: Early Stopping

Regularización: Early Stopping

Train Test split:



Regularización: Early Stopping

Parar el entrenamiento antes de que haya overfitting



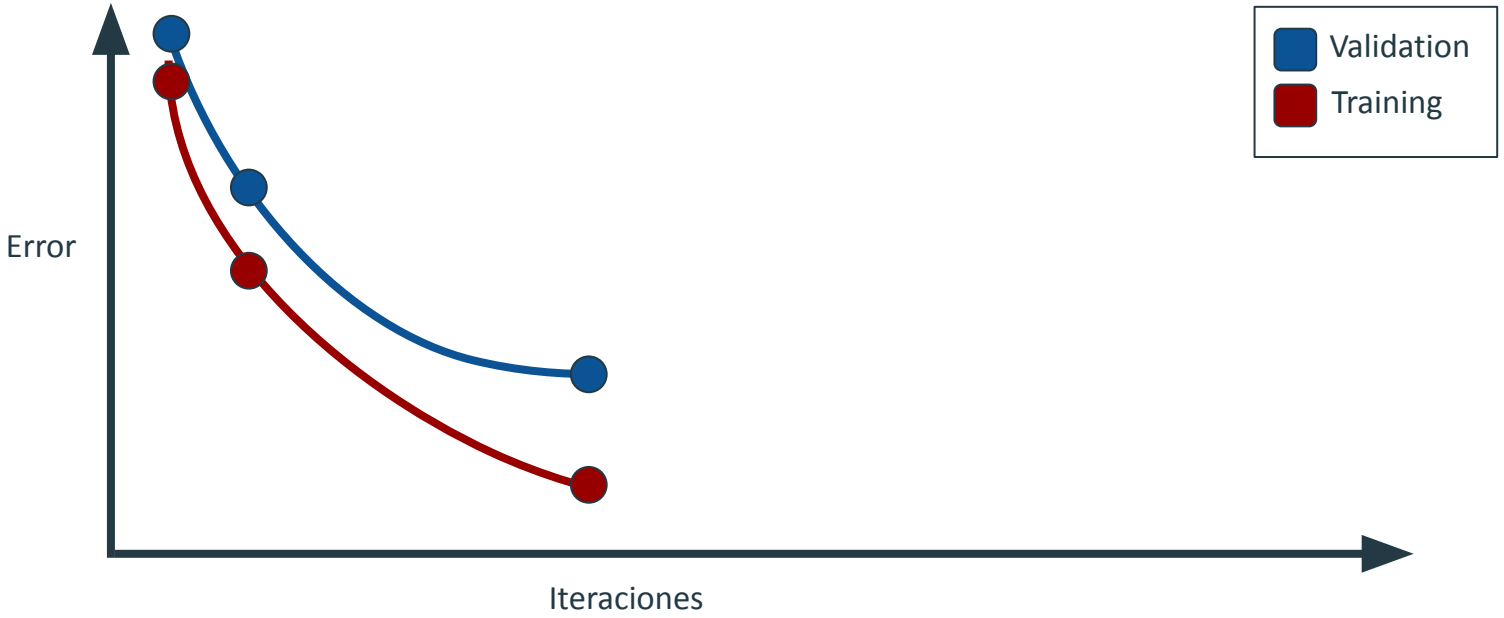
Regularización: Early Stopping

Parar el entrenamiento antes de que haya overfitting



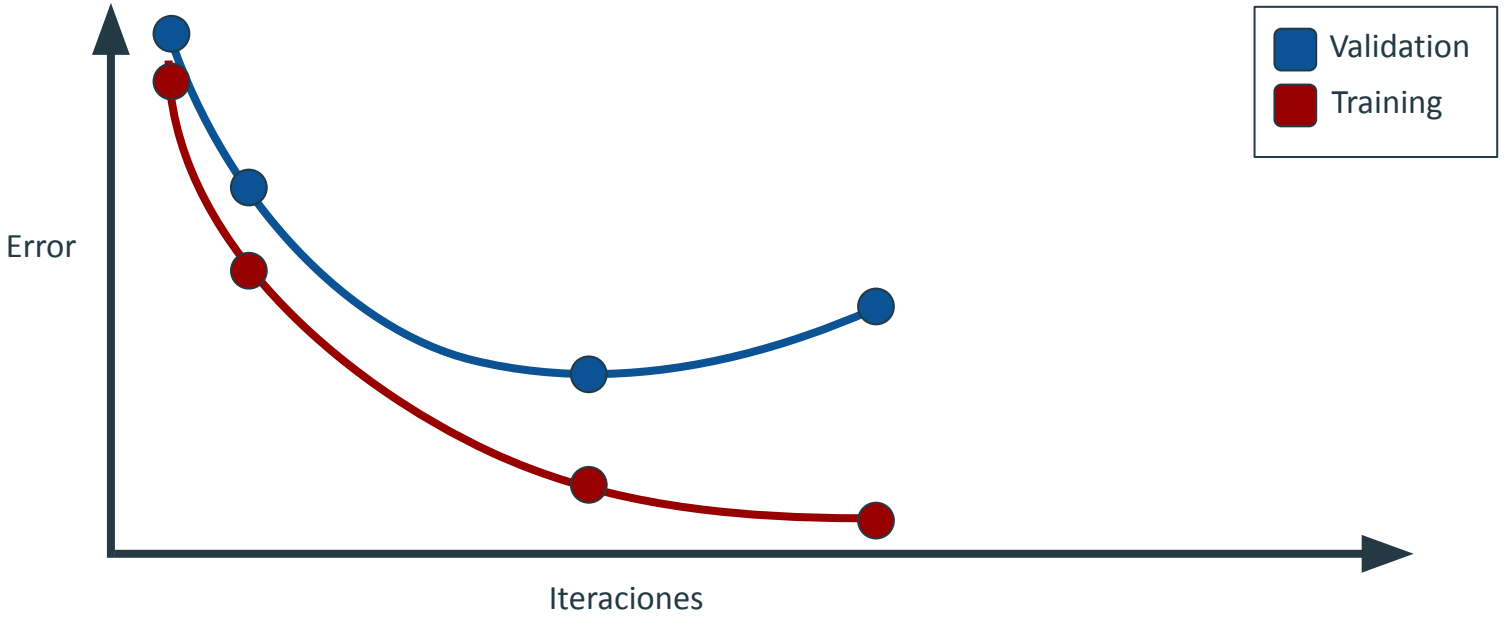
Regularización: Early Stopping

Parar el entrenamiento antes de que haya overfitting



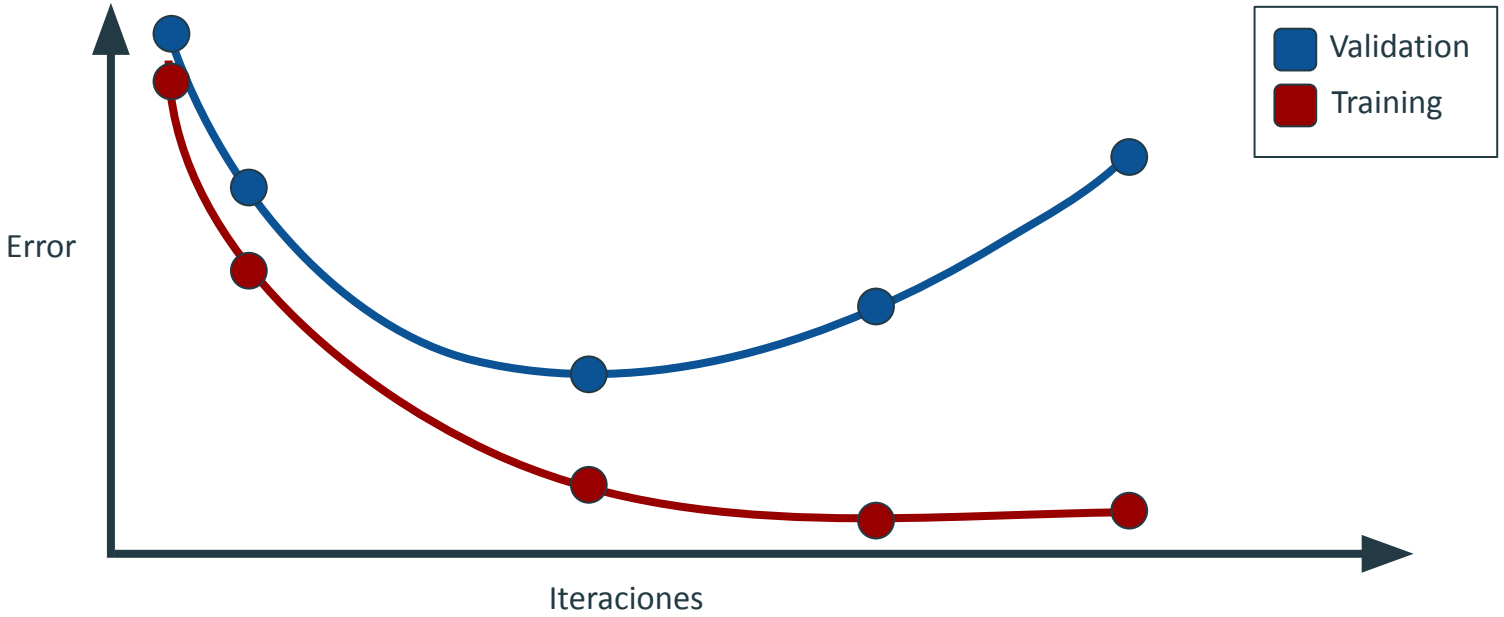
Regularización: Early Stopping

Parar el entrenamiento antes de que haya overfitting



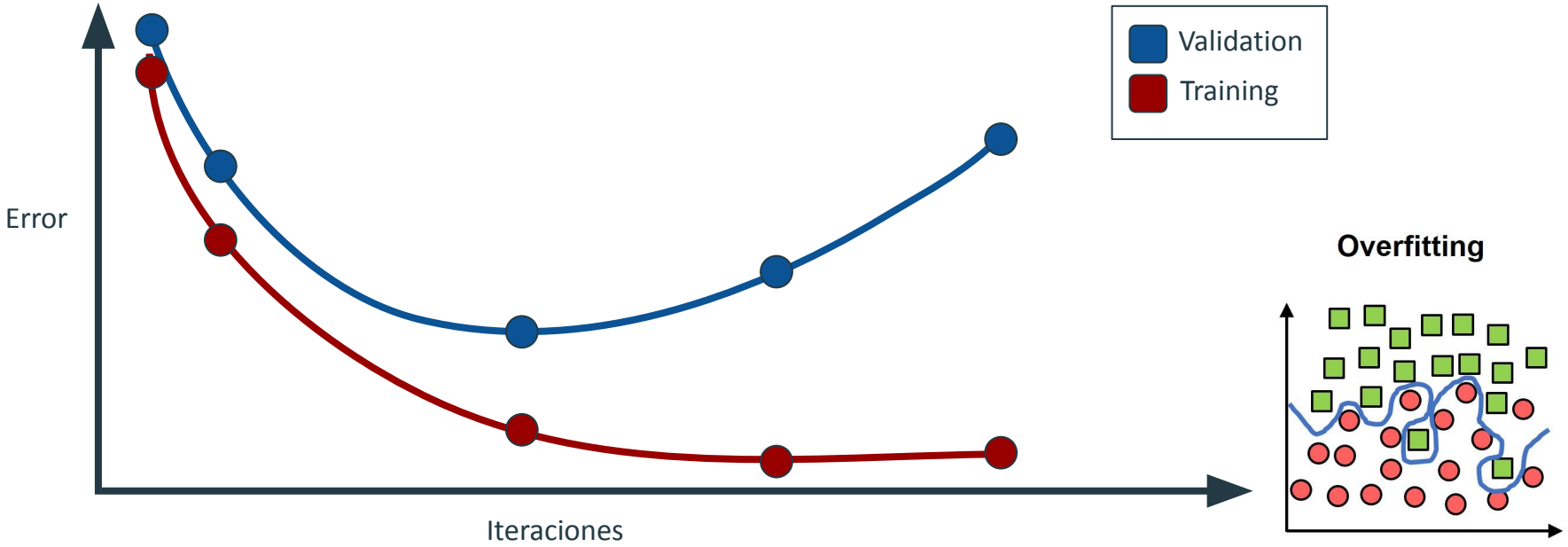
Regularización: Early Stopping

Parar el entrenamiento antes de que haya overfitting



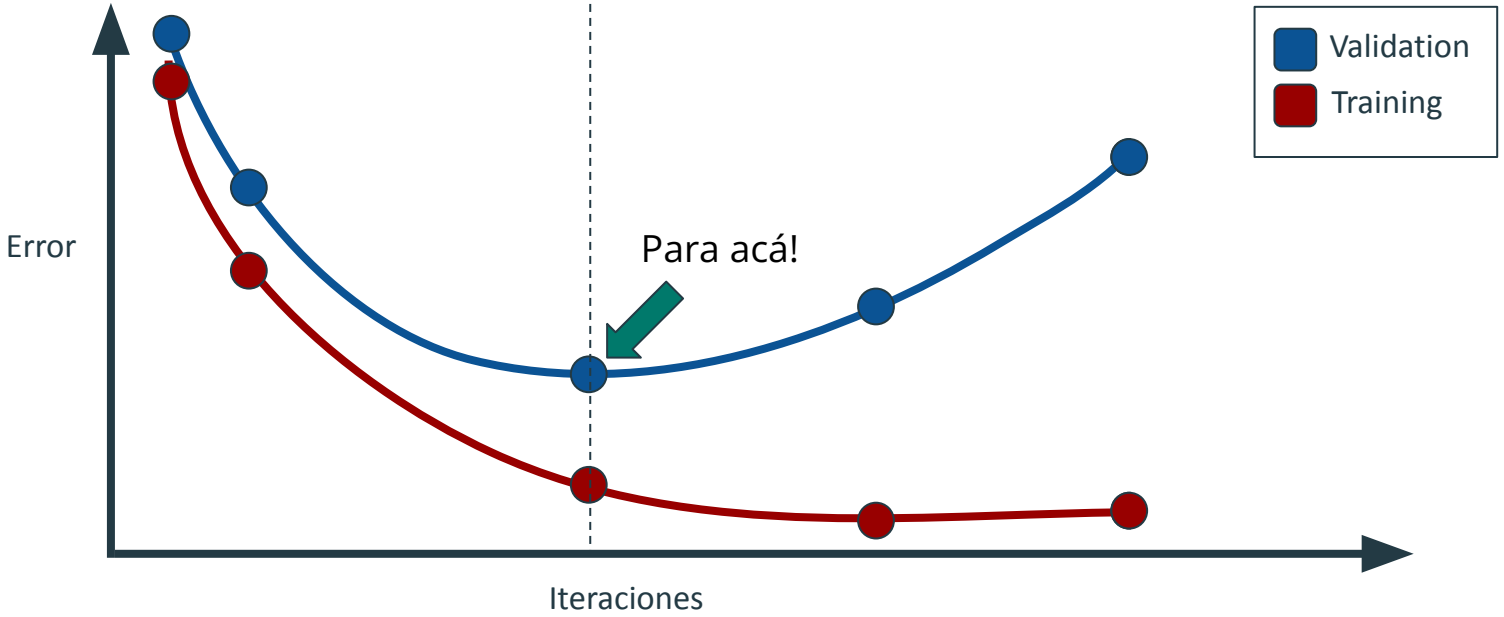
Regularización: Early Stopping

Parar el entrenamiento antes de que haya overfitting

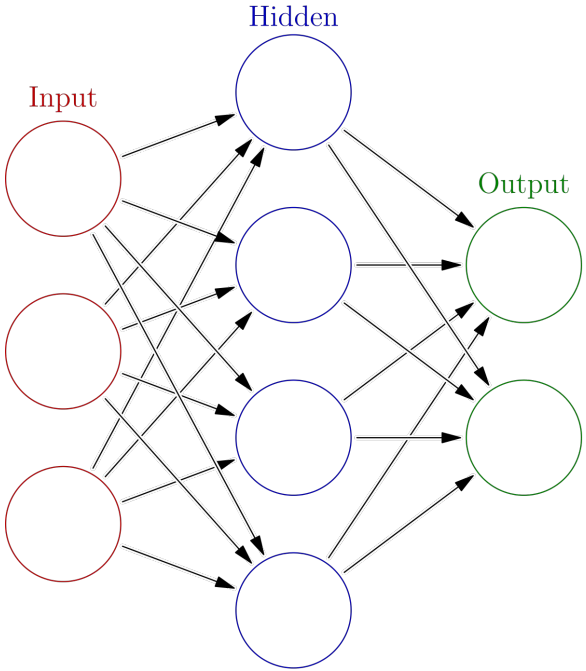


Regularización: Early Stopping

Parar el entrenamiento antes de que haya overfitting



Red Neuronal



Un perceptrón simple es equivalente a una regresión logística.

Link

Qué vimos hoy

- Regresión lineal
- Perceptrón simple
- Red Neuronal Profunda
- Backpropagation
- Learning Rate
- Funciones de activación
- Regularización:
  - Dropout
  - Early Stopping